```
# 1. IMPORT LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from textblob import TextBlob
from wordcloud import WordCloud, STOPWORDS


# 2. LOAD DATASET
# ========================
# Make sure file name is correct (NO extra space)
df = pd.read_csv("Sentiment_Analysis .csv")

# View data
print(df.head())


# 3. SENTIMENT FUNCTION

def get_sentiment(text):
    meaning = TextBlob(text)
    if meaning.sentiment.polarity > 0:
        return "Positive"
    elif meaning.sentiment.polarity < 0:
        return "Negative"
    else:
        return "Neutral"

# Apply sentiment analysis
df["Sentiment"] = df["Sentences"].apply(get_sentiment)

print(df.head())


# 4. SAVE OUTPUT

df.to_excel("Sentiment_Labeled.xlsx", index=False)


# 5. SENTIMENT COUNT

sentiment_count = df["Sentiment"].value_counts()

print(sentiment_count)

# 6. PIE CHART

plt.figure(figsize=(5,5))
plt.pie(sentiment_count, labels=sentiment_count.index, autopct="%1.2f%%")
plt.title("Sentiment Distribution (Pie Chart)")
plt.show()


# 7. BAR CHART

plt.figure()
sentiment_count.plot(kind='bar')
plt.title("Sentiment Distribution (Bar Chart)")
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.show()

# 8. BAR CHART

plt.figure()
sns.countplot(x='Sentiment', data=df)
plt.title("Sentiment Distribution (Seaborn)")
plt.show()


# 9. FILTER NEGATIVE REVIEWS

negative_reviews = df[df["Sentiment"] == "Negative"]

print(negative_reviews.head())

# Combine all negative text
negative_text = " ".join(negative_reviews["Sentences"])
```

```
negative_text = " ".join(negative_reviews['Sentences'])


# 10. WORD CLOUD

stopwords = set(STOPWORDS)
stopwords.update(["product", "service"])  # optional

wc = WordCloud(
    width=800,
    height=400,
    background_color='black',
    stopwords=stopwords
).generate(negative_text)

plt.figure()
plt.imshow(wc)
plt.axis("off")
plt.title("Most Common Words in Negative Reviews")
plt.show()
```

```
                                           Sentences
0  The product's exceptional quality and performa...
1  Poor customer support and the product's subpar...
2  The service met standard requirements without ...
3  Outstanding customer service accompanied the p...
4  The product's failure to meet basic requiremen...
                                           Sentences Sentiment
0  The product's exceptional quality and performa...  Positive
1  Poor customer support and the product's subpar...  Negative
2  The service met standard requirements without ...  Positive
3  Outstanding customer service accompanied the p...  Positive
4  The product's failure to meet basic requiremen...  Negative
Sentiment
Positive    12
Negative     9
Neutral      1
Name: count, dtype: int64
```